The 16-bit FAT filesystem

WHAT IS THIS FILESYSTEM?

FAT16 uses 16-bit allocation entries to link file clusters. It is familiar from DOS-era disks, older memory cards and removable drives.

How the disk is organised

Simplified illustration; not a sector-for-sector specification.

What this implementation provides

- 16-bit allocation table and fixed root-directory layout
- Data-cluster range 4,085 to 65,524
- V3 file/directory operations; V4 formatting and metadata; V5 dates and labels
- Generic Boot bridge and Defragmenter consume the same FAT16.so

One installed module and one manual

```
/System/Filesystems/FAT16.so
/System/Resources/Documentation/Filesystem-FAT16/
FAT16.pdf
```

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How to use the installed module

The FAT16.so driver is a shared library. It is loaded by a compatible filesystem consumer through its published versioned interfaces; it is not a command to run directly.

Installation and verification

Portable package: install the released Debian .deb.

Native option: run the released .run installer.

The Debian package owns exactly one .so and this PDF.

Build records and checksums live in the Debian package metadata.
Full source, tests and engineering notes remain in GitHub.

Current limits and boundaries

- FAT12 and FAT32 are not interchangeable with FAT16.
- A Linux kernel module and an EFI driver are separate transitional builds.
- Do not treat test coverage as physical-media or power-loss certification.

Operational model

```
Media -> host I/O callbacks -> filesystem .so -> caller
```

The host supplies target selection and protected byte I/O. The filesystem driver interprets that filesystem’s on-disk structures. Ownership, verification and all destructive actions remain the responsibility of the calling application.

This is a user guide, not a claim of untested kernel or firmware support.

Build and installation

Filesystem-FAT16

Release 0.1.10

Installed files

```
/System/Filesystems/FAT16.so

/System/Resources/Documentation/Filesystem-FAT16/FAT16.pdf

/System/Kernel/Filesystems/fat16.ko
```

Debian package

Install the prebuilt files with dpkg -i followed by the Debian filename.

Manual RUN

Run bash followed by the native installer filename.

The installer compiles the shared object and generates this manual locally.

It also compiles the Linux adapter against the running kernel headers.

Use --kernel-build DIRECTORY to select another prepared kernel build.

The manual installer replaces all three files.

InfiltratorOS installation

The OS installer builds a missing or older shared object.

It verifies the manual independently and repairs it from its matching release.

The kernel module is stored for manual use under its single filename.

The package metadata records the source revision and SHA-256 checksums.